1.2.2 Running a Java EE application in your private network

Maureen is a senior system architect in a global corporation. She wants to move parts of her company's business applications to AWS when the data center contract expires in a few months, to reduce costs and gain flexibility. She would like to run enterprise applications (such as Java Enterprise Edition [EE] applications) consisting of an application server and an SQL database on AWS. To do so, she defines a virtual network in the cloud and connects it to the corporate network through a virtual private network (VPN) connection. She installs application servers on virtual machines to run the Java EE application. Maureen also wants to store data in an SQL database service (such as Oracle Database EE or Microsoft SQL Server EE).

For security, Maureen uses subnets to separate systems with different security levels from each other. By using access-control lists, she can control ingoing and outgoing traffic for each subnet. For example, the database is accessible only from the Java EE server's subnet, which helps to protect mission-critical data. Maureen controls traffic to the internet by using network address translation (NAT) and firewall rules as well. Figure 1.4 illustrates Maureen's architecture.